

Computational methods and their limits

For the two matched-pair experiments on DASA open/closed thermodynamics. Written 2026-09-09. Every number here is reproducible from `notebooks/data/xtb_cache.sqlite` via `notebooks/dft_ladder.py` and `notebooks/dft_molecule.py`.

1. The quantity

For each molecule, $\Delta E = E(\text{closed}) - E(\text{open})$, where $E(\text{closed})$ is a Boltzmann sum at 298.15 K over the diastereomers of the **neutral keto** closed form. Results are reported only as **$\Delta\Delta E$ between a matched pair**, never as absolute values.

Two justifications for the choices, both from the literature rather than convenience:

- **Keto only.** Lerch, Feringa, Buma et al., *Angew. Chem. Int. Ed.* **2018**, 57, 8063 (Fig. 1c): in aprotic solvents the cyclization stops at the neutral form; the zwitterion is the polar-protic product. Peterson, Stricker & Read de Alaniz, *Chem. Commun.* **2022**, 58, 2303 likewise define ΔG_{CA} against a "closed **keto** form (C)". Including the zwitterion in a toluene calculation is wrong on the literature's own account, and empirically it destroyed the ranking against nine measured CDCl_3 equilibria (Spearman $\rho +0.353 \rightarrow +0.748$ on removing it).
- **Differences only.** Absolute ΔE is not ΔG and is not close to it: for D1-H, computed $\Delta E = -6.48$ against a measured $\Delta G = -0.53$ kcal/mol (29% open), a **-5.95 kcal/mol** systematic offset. Only matched-pair differences are used.

2. Protocol

Identical for all five molecules:

functional	M06-2X — the functional used by Peterson for these systems
geometries	GFN2-xTB , ALPB(toluene); open form = flattest of 120 ETKDG/MMFF conformers, closed = lowest-MMFF of 20
solvation	C-PCM , $\epsilon = 2.374$ (toluene)
energies	electronic ; no ZPE, no thermal correction, no entropy
Coulomb	density fitting
closed manifold	Boltzmann sum, 298.15 K

Differs between the two experiments — and this is the main caveat:

	basis	grid	basis fns	unique closed states computed
Ladder (D1-H, D1-M)	6-31+G(d,p)	3	545 / 573	2 of 2 — complete
Acceptor bulk (3-Me, 3-iPr, 3-pentyl)	6-31G(d)	2	488 / 560	4 of 4 — complete

Within each experiment the protocol is **identical on both sides**, so each $\Delta\Delta E$ is a like-for-like difference. Across experiments it is not.

3. Why single points on semiempirical geometries

Not an assumption — it was tested. Holding geometries fixed and changing only the electronic structure method:

	$\Delta\Delta E$ for a C5 methyl	error
GFN2-xTB	−1.39 (wrong sign)	−3.66
M06-2X // same xTB geometries	+0.89	−1.39
measured (29% → 95% open)	+2.275	—

So the functional, not the geometry, was the limiting approximation. A DFT geometry optimisation was measured at **5–12 h per species** on this hardware (M06-2X/6-31G(d), ~40 atoms, 8 threads) against ~25–45 min for a single point, which is why it was not done. The residual −1.39 kcal/mol means the method recovers ~40% of a known steric effect: **useful for sign and ordering, not for magnitude**.

4. Known approximations, with expected sign

approximation	expected effect on a steric $\Delta\Delta E$
electronic energy, not free energy	ring closure restricts rotors; the bulkier partner loses more → true $\Delta\Delta G$ more positive
xTB geometries	GFN2 is soft on short-range repulsion → strain under-relaxed → understates
intramolecular BSSE	more basis functions on the larger partner stabilise its compact closed form → understates
C-PCM instead of SMD	identical on both sides of a matched pair → cancels
3-alkyl is more electron-donating than methyl	weaker acceptor → less charge separation → less open → opposes the observed sign

Every identified systematic runs the same way: the measured effect is a **lower bound**.

The last row is why the isopropyl/3-pentyl pair is the comparison to quote: both are secondary CHR₂ carbons, so the electronic term is matched between them and only bulk varies. It is *not* matched between methyl and either of them, and that is exactly where the ladder breaks (§5).

The noise floor, measured for free

A closed DASA has no external chiral reference, so a stereoisomer and its mirror image are the same physical species and **must** be exactly degenerate. RDKit enumerates both, which gives an internal control that costs nothing: any energy gap between an enantiomeric pair is pure conformer-search error.

Measured on 3-methyl at M06-2X: **0.69 kcal/mol**. That is the uncertainty on every $\Delta\Delta E$ in §5, and the bar a claimed effect must clear. $\Delta\Delta E(\text{pentyl} - \text{isopropyl}) = +4.27$ clears it by 6.2 $\times$; the confounded +2.12 clears it by 3.1 $\times$; an effect of ~ 0.5 would not clear it at all.

Enantiomer duplicates are now collapsed at enumeration time (`_mirror_image` in `dasa_chem.py`, sentinel group [13]), which both removes the $-RT \ln 2 = -0.41$ kcal/mol double-counting bias and halves the manifold cost. The collapse is lossless: each discarded structure is recorded as its representative's `mirror`, so cached energies remain findable under either label.

5. Results

Experiment A — triene C5 methyl (validation). D1-H vs D1-M, indoline donor, Meldrum's acid acceptor, complete manifolds, full recipe. $\Delta\Delta E = +0.89$ against a measured **+2.275** kcal/mol. Correct sign; 2.6 $\times$ better than GFN2-xTB on identical geometries.

Experiment B — acceptor 3-position bulk (new). 1-phenylpyrazol-5-one, N-pentenylaniline donor, reduced recipe, **complete 4-of-4 manifolds on all three legs** (2026-09-09). Identical donor, acceptor ring, triene, solvent.

C3 substituent	class	states	ΔE (kcal/mol)
methyl, CH ₃	primary	4/4	-8.54
isopropyl, CH(CH ₃) ₂	secondary	4/4	-10.69
3-pentyl, CH(CH ₂ CH ₃) ₂	secondary	4/4	-6.42

The ladder is **not monotonic in size**: isopropyl stabilises the closed form *more* than methyl. The reason is that methyl→isopropyl changes the substitution **class** (primary→secondary), moving electronics and sterics together, while isopropyl→3-pentyl holds the class fixed and varies only bulk. Two comparisons therefore exist and only one is clean:

- $\Delta\Delta E(\text{pentyl} - \text{methyl}) = +2.12$ — confounded, electronics and sterics mixed. *This supersedes the +1.95 previously reported here, which additionally double-counted enantiomers.*

- $\Delta\Delta E(\text{pentyl} - \text{isopropyl}) = +4.27$ — electronically matched pair, **only bulk differs**. $6.2\times$ the 0.69 kcal/mol noise floor. **This is the result.**

Acceptor bulk destabilises the closed form, and by roughly twice the magnitude the confounded comparison suggested. Methyl was simply the wrong reference.

Convergence. Completing the manifolds changed ΔE for isopropyl and 3-pentyl by **0.00 kcal/mol** — the states added last carry $<0.005\%$ Boltzmann weight. The sums are dominated by the lowest one or two states. This could only be known by computing them, and it retires weakness 1 below.

This is the design axis with no literature precedent: Meldrum's acid and barbituric acid cannot carry a substituent adjacent to the acidic carbon (both flanking positions are carbonyls), whereas pyrazolones and isoxazolones can.

6. What is NOT defensible, stated plainly

- **Comparing a raw ΔE across the two experiments.** An earlier note observed that 3-pentyl (−6.42) and D1-H (−6.48) were "essentially identical". That comparison spans two basis sets, two grids and two manifold completenesses. **Withdrawn.**
- **Offset-correcting Experiment B with the D1-H offset** (−5.95, derived at the full recipe) to estimate "% open". That transfers a recipe-specific constant across recipes *and* across a donor and acceptor-class change. **Withdrawn;** the ~31%-open figure it produced should not be used.
- **Extrapolating +1.95 by the ladder's ~40% recovery factor.** That factor was measured for a C5-triene substituent at the full recipe. Suggestive, not a result.

7. Residual weaknesses, ranked

1. ~~Incomplete manifolds in Experiment B.~~ **RESOLVED 2026-09-09.** All three legs now carry complete 4-of-4 manifolds. The completion changed the two secondary-alkyl ΔE values by 0.00 kcal/mol, so the Boltzmann sums were already converged — but this was not knowable in advance, and the methyl completion is what revealed the non-monotonicity as real rather than a coverage artifact.
2. ~~Diastereomer selection uses a 2.5 kcal/mol xTB cutoff.~~ **RESOLVED, and the underlying assumption was found to be false.** Every unique state is now computed at DFT, so no xTB-based selection remains anywhere in Experiment B. The reason this mattered: measured over all three molecules, GFN2-xTB's ranking of closed-form diastereomers is **uncorrelated with M06-2X** — Kendall $\tau = -0.061$, Spearman $\rho = -0.086$, 5 of 9 pairwise orderings correct against a 50 % coin flip, and xTB picked the wrong DFT minimum in **3 of 3** molecules. On 3-isopropyl xTB compressed a genuine 7.71 kcal/mol spread into 1.48. **Selecting DFT subsets by xTB energy is therefore invalid for this system.** Experiment A was never exposed to this — `dft_ladder.units()` enumerates exhaustively and both molecules are complete at 2/2 — but any future campaign must enumerate rather than pre-screen.

3. **Geometries are GFN2-xTB throughout.** Weakness 2 establishes that xTB *energies* are unreliable for these diastereomers; it does not by itself establish that its *geometries* are, and M06-2X single points on them recover the sign in Experiment A. But the question is now open for the bulky cases the steric result depends on. **Fix: M06-2X geometry optimisation — hours per structure ×10.** This is the largest remaining exposure and the most expensive to close.
4. **No free-energy correction.** Expected to increase the effect (§4), not reverse it, but unquantified.
5. **The two experiments use different basis sets and grids.** Each $\Delta\Delta E$ is internally consistent; they are not quantitatively comparable to each other. **Fix: run D1-H and D1-M at the reduced recipe (~3 h) to measure the recipe's effect on a $\Delta\Delta E$.**

8. An error found and corrected

`closed_stereoisomers` enumerated **enantiomers as distinct species**. Enantiomers are degenerate, so including both double-counts a physical state and artificially lowers a Boltzmann sum. For 3-methyl, one of the three structures sent to DFT was the mirror image of another. Removing it moved **$\Delta\Delta E$ from +2.10 to +1.95 kcal/mol**; the value in §5 is the corrected one.

The enantiomer energy gap — which must be exactly zero — is retained as a free convergence diagnostic. Measured gaps of 0.08–6.2 kcal/mol are the evidence behind weakness 2 above.